

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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Linear phase FIR filter

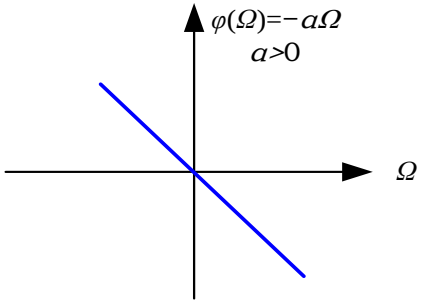
- Sufficient and necessary conditions for linear phase systems
- Time domain characteristics of linear phase systems
- Frequency domain characteristics of linear phase systems
- Zero distribution of linear phase system

Sufficient and necessary conditions for linear phase systems

Strictly Linear Phase System

$$H(e^{j\Omega}) = \left|H(e^{j\Omega})\right| e^{j\varphi(\Omega)}$$

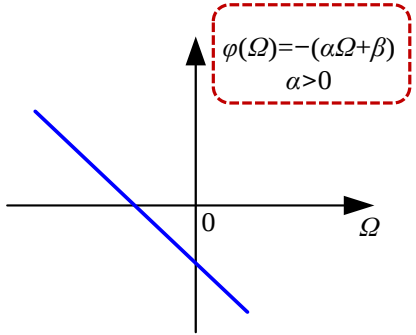
$$\varphi(\Omega) = -a\Omega$$



Generalized Linear Phase System

$$H(e^{j\Omega}) = A(\Omega)e^{-j(\alpha\Omega+\beta)}$$

α and β are constants that are independent of Ω, and $A(\Omega)$ is a real function that can be either positive or negative.



Sufficient and necessary conditions for linear phase systems

$$H(z)=\sum_{k=0}^{M} h[k]z^{-k} \qquad N=M+1$$

M is the order of the FIR digital filter, and N is the length of h[k].

The necessary and sufficient condition for an FIR digital filter to have a linear phase is

Time domain: $h[k] = \pm h[M-k]$

z-domain: $H[z] = \pm z^{-M} H[z^{-1}]$

Sufficient and necessary conditions for linear phase systems

$$h[k] = h[M-k]$$

$h[k]=\{2, 3, 5, 3, 2\}, \quad M=4$

$$H(z) = 2 + 3z^{-1} + 5z^{-2} + 3z^{-3} + 2z^{-4}$$

$$H(z^{-1}) = 2 + 3z + 5z^{2} + 3z^{3} + 2z^{4}$$

$$z^{-4}H(z^{-1}) = 2z^{-4} + 3z^{-3} + 5z^{-2} + 3z^{-1} + 2$$

$$H(z) = z^{-4}H(z^{-1})$$

z-domain： $H[z] = z^{-M}H[z^{-1}]$

$$h[k] = -h[M-k]$$

$h[k]=\{2, 3, 0, -3, -2\}, \quad M=4$

$$H(z) = 2 + 3z^{-1} - 3z^{-3} - 2z^{-4}$$

$$H(z^{-1}) = 2 + 3z - 3z^{3} - 2z^{4}$$

$$z^{-4}H(z^{-1}) = 2z^{-4} + 3z^{-3} - 3z^{-1} - 2$$

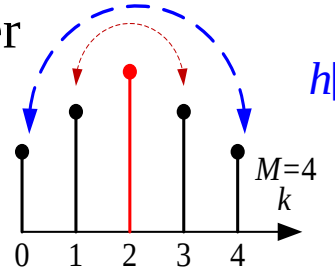
$$H(z) = -z^{-4}H(z^{-1})$$

z-domain： $H[z] = -z^{-M}H[z^{-1}]$

Time domain characteristics of linear phase systems

Type I Linear Phase System

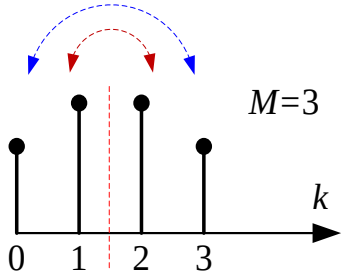
$h[k]$ even symmetry， M is an even number



$$h[k]=\{\underset{\downarrow}{2},3,4,3,2\}$$

Type II Linear Phase System

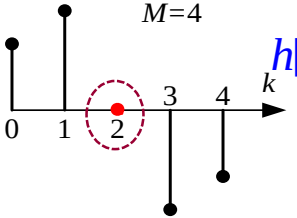
$h[k]$ even symmetry ， M is an odd number



$$h[k]=\{\underset{\downarrow}{2},3,3,2\}$$

Type III Linear Phase System

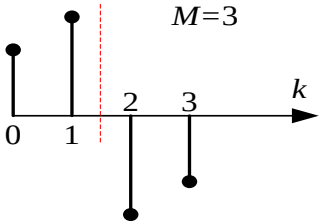
$h[k]$ odd symmetry ， M is an even number



$$h[k]=\{\underset{\downarrow}{2},3,0,-3,-2\}$$

Type IV Linear Phase System

$h[k]$ odd symmetry， M is an odd number



$$h[k]=\{\underset{\downarrow}{2},3,-3,-2\}$$

Time domain characteristics of linear phase systems

※ Tpye I ($h[k]=h[M-k]$, M is an even number.)

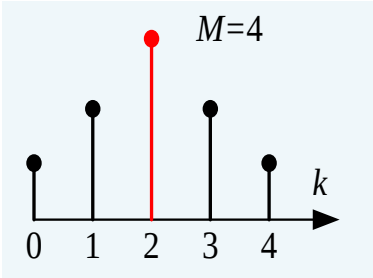
※ Tpye II ($h[k]=h[M-k]$, M is an odd number.)

※ Type III ($h[k]=-h[M-k]$, M is an even number.)

※ Type IV($h[k]=-h[M-k]$, M is an odd number.)

Frequency domain characteristics of linear phase systems

Type I ($h[k]=h[M-k]$, M is even number)



$$h[k]=\{1,2,3,2,1\} \qquad M=4$$

$$H(\mathrm{e}^{\mathrm{j}\Omega})=1+2\mathrm{e}^{-\mathrm{j}\Omega}+3\mathrm{e}^{-\mathrm{j}2\Omega}+2\mathrm{e}^{-3\mathrm{j}\Omega}+1\cdot\mathrm{e}^{-4\mathrm{j}\Omega}$$

$$=\mathrm{e}^{-2\mathrm{j}\Omega}(1\cdot\mathrm{e}^{\mathrm{j}2\Omega}+1\cdot\mathrm{e}^{-\mathrm{j}2\Omega})+\mathrm{e}^{-\mathrm{j}2\Omega}(2\mathrm{e}^{\mathrm{j}\Omega}+2\mathrm{e}^{-\mathrm{j}\Omega})+3\mathrm{e}^{-\mathrm{j}2\Omega}$$

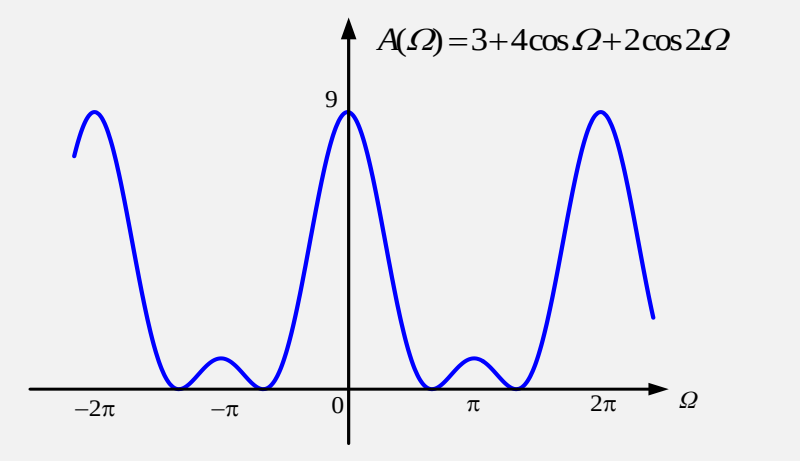
$$=\mathrm{e}^{-\mathrm{j}2\Omega}(3+4\cos\Omega+2\cos 2\Omega)$$

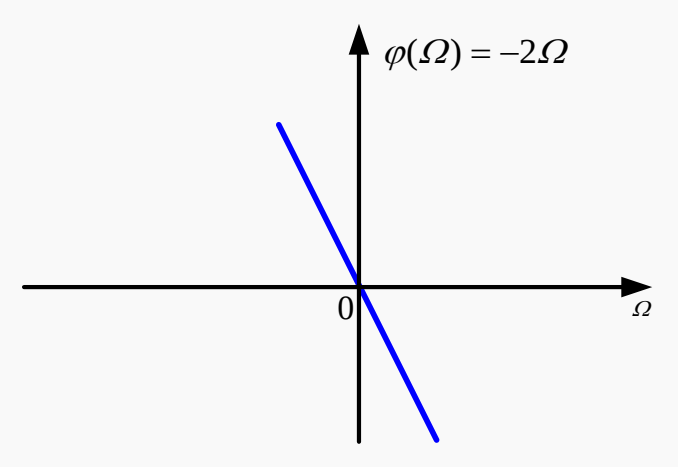
$$A(\Omega)=3+4\cos\Omega+2\cos 2\Omega \qquad \varphi(\Omega)=-2\Omega=-\frac{M}{2}\Omega$$

Frequency domain characteristics of linear phase systems

Type I ($h[k]=h[M-k]$, M is even number)

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{e}^{-\mathrm{j}2\Omega}(3+4\cos\Omega+2\cos 2\Omega)$$



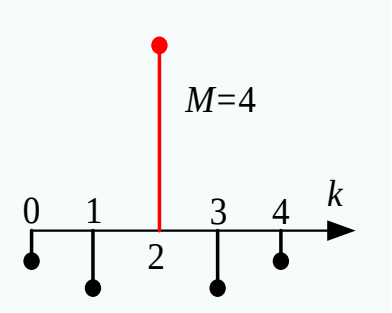


$A(-\Omega) = A(\Omega)$ $A(\Omega)$ is even-symmetric about the point $\Omega = 0$

$A(2\pi - \Omega) = A(\Omega)$ $A(\Omega)$ is odd-symmetric about the point $\Omega = \pi$

Frequency domain characteristics of linear phase systems

Type I ($h[k]=h[M-k]$, M is even number)



$$h[k]=\{-1,-2,6,-2,-1\} \qquad M=4$$

$$H(e^{j\Omega})=-1-2e^{-j\Omega}+6e^{-j2\Omega}-2e^{-j3\Omega}-1\cdot e^{-j4\Omega}$$

$$=e^{-j2\Omega}(-1\cdot e^{j2\Omega}-1\cdot e^{-j2\Omega})+e^{-j2\Omega}(-2e^{j\Omega}-2e^{-j\Omega})+6e^{-j2\Omega}$$

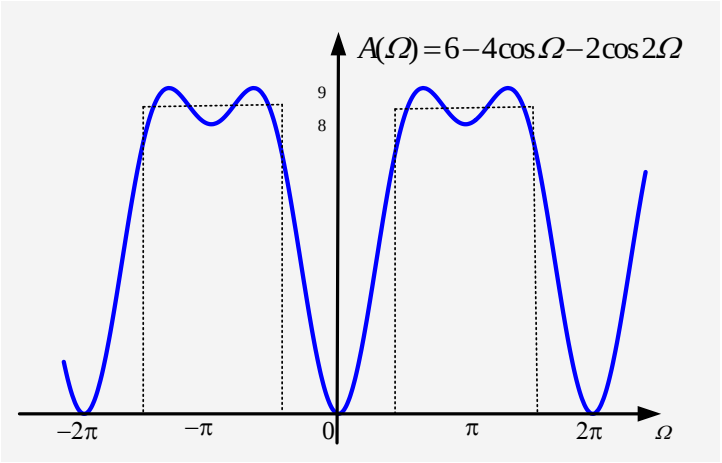
$$=e^{-j2\Omega}(6-4\cos\Omega-2\cos 2\Omega)$$

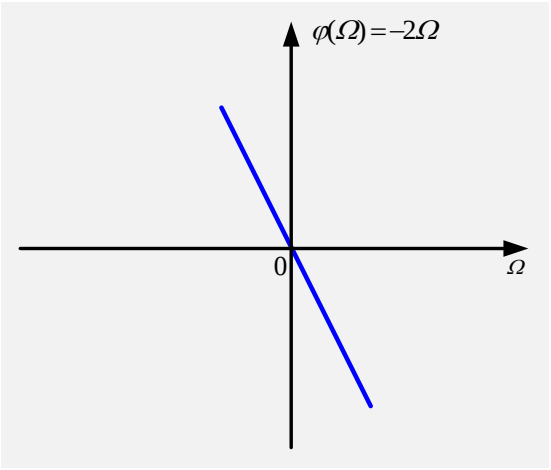
$$A(\Omega)=6-4\cos\Omega-2\cos 2\Omega \qquad \varphi(\Omega)=-2\Omega=-\frac{M}{2}\Omega$$

Frequency domain characteristics of linear phase systems

Type I ($h[k]=h[M-k]$, M is even number)

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{e}^{-\mathrm{j}2\Omega}(6-4\cos\Omega-2\cos 2\Omega)$$





$$A(-\Omega) = A(\Omega)$$

$$A(2\pi-\Omega) = A(\Omega)$$

$A(\Omega)$ is even-symmetric about the point $\Omega = 0$

$A(\Omega)$ is even-symmetric about the point $\Omega = \pi$

Frequency domain characteristics of linear phase systems

Type I ($h[k]=h[M-k]$, M is even number)

$$H\left(e^{j\Omega}\right)=e^{-j0.5M\Omega}A(\Omega)$$

$$\varphi(\Omega)=-\frac{M}{2}\Omega$$

$A(\Omega)$ is even-symmetric about the point $\Omega=0$

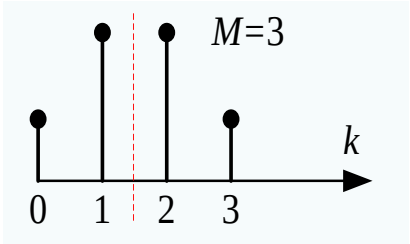
$A(\Omega)$ is even-symmetric about the point $\Omega=\pi$

Can design LP、HP、BP、BS

Frequency domain characteristics of linear phase systems

Type II ($h[k]=h[M-k]$, M is odd number)

$h[k]=\{1,2,2,1\}$ $\quad M=3$



$$H(\mathrm{e}^{\mathrm{j}\Omega})=1+2\mathrm{e}^{-\mathrm{j}\Omega}+2\mathrm{e}^{-\mathrm{j}2\Omega}+1\cdot\mathrm{e}^{-3\mathrm{j}\Omega}$$

$$=\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}(\mathrm{e}^{\mathrm{j}\frac{3}{2}\Omega}+\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega})+\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}(2\mathrm{e}^{\mathrm{j}\frac{\Omega}{2}}+2\mathrm{e}^{-\mathrm{j}\frac{\Omega}{2}})$$

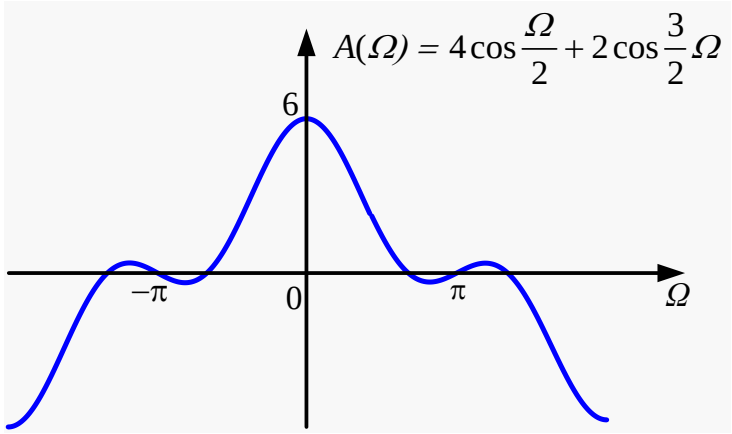
$$=\mathrm{e}^{-\mathrm{j}\frac{3}{2}\Omega}(4\cos\frac{\Omega}{2}+2\cos\frac{3}{2}\Omega)$$

$$A(\Omega)=4\cos\frac{\Omega}{2}+2\cos\frac{3}{2}\Omega \qquad \varphi(\Omega)=-\frac{3}{2}\Omega=-\frac{M}{2}\Omega$$

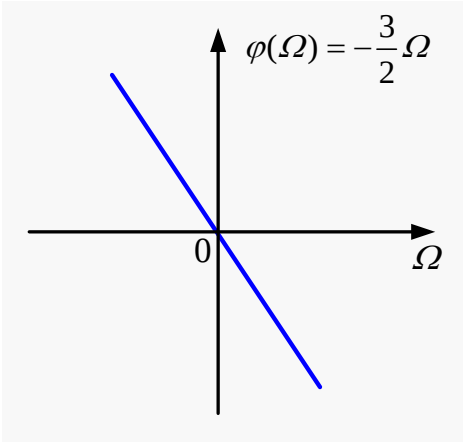
Frequency domain characteristics of linear phase systems

Type II ($h[k]=h[M-k]$, M is odd number)

$$H(e^{j\Omega}) = e^{-j\frac{3}{2}\Omega}\left(4\cos\frac{\Omega}{2} + 2\cos\frac{3}{2}\Omega\right)$$



$$A(\pi) = 0$$



$A(-\Omega) = A(\Omega)$ $A(\Omega)$ is even-symmetric about the point $\Omega = 0$

$A(2\pi - \Omega) = -A(\Omega)$ $A(\Omega)$ is odd-symmetric about the point $\Omega = \pi$

Frequency domain characteristics of linear phase systems

Type II ($h[k]=h[M-k]$, M is odd number)

$$H(e^{j\Omega}) = e^{-j0.5M\Omega} A(\Omega)$$

$$\varphi(\Omega) = -\frac{M}{2}\Omega$$

$A(\Omega)$ is even-symmetric about the point $\Omega = 0$

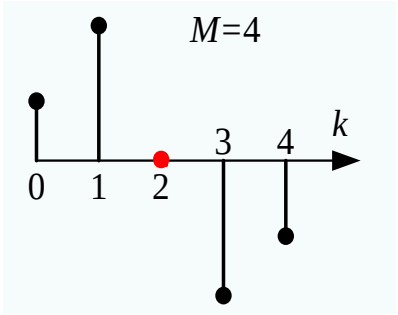
$A(\Omega)$ is even-symmetric about the point $\Omega = \pi$ $\quad A(\pi) = 0$

Cannot be used for the design of high-pass or band-stop filters.

Frequency domain characteristics of linear phase systems

Type III ($h[k]= -h[M-k]$, M is even number)

$h[k]=\{1, 2, 0, -2, -1\}$, $M=4$



$$H(e^{j\Omega}) = 1 + 2e^{-j\Omega} - 2e^{-j3\Omega} - 1\cdot e^{-j4\Omega}$$

$$= e^{-j2\Omega}(e^{j2\Omega} - e^{-j2\Omega}) + e^{-j2\Omega}(2e^{j\Omega} - 2e^{-j\Omega})$$

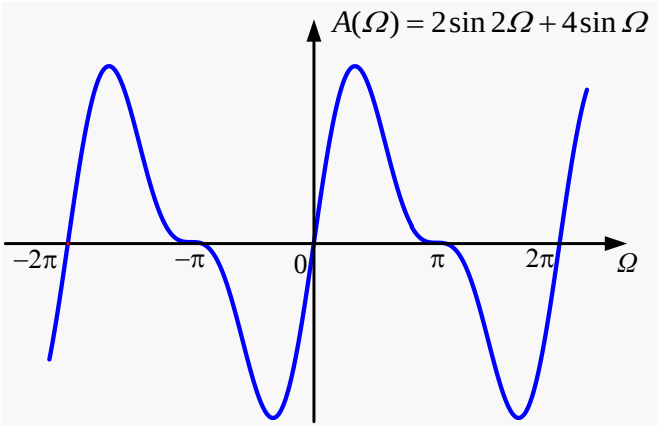
$$= e^{-j(2\Omega-\pi/2)}(2\sin 2\Omega + 4\sin\Omega)$$

$$A(\Omega) = 2\sin 2\Omega + 4\sin\Omega \qquad \varphi(\Omega) = -2\Omega + \frac{\pi}{2} = -\frac{M}{2}\Omega + \frac{\pi}{2}$$

Frequency domain characteristics of linear phase systems

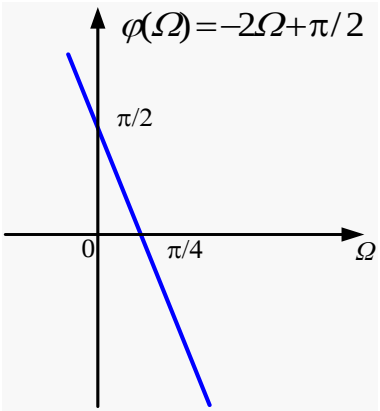
Type III ($h[k]= -h[M-k]$, M is even number)

$$H(e^{j\Omega})=e^{-j(2\Omega-\pi/2)}(2\sin 2\Omega+4\sin\Omega)=e^{-j(0.5M\Omega-0.5\pi)}A(\Omega)$$



$A(0)=0$

$A(\pi)=0$



$A(-\Omega)=-A(\Omega)$

$A(2\pi-\Omega)=-A(\Omega)$

$A(\Omega)$ is odd-symmetric about the point $\Omega=0$

$A(\Omega)$ is odd-symmetric about the point $\Omega=\pi$

Frequency domain characteristics of linear phase systems

Type III ($h[k]= -h[M-k]$, M is even number)

$$H(\mathrm{e}^{\mathrm{j}\Omega}) = \mathrm{e}^{-\mathrm{j}(0.5M\Omega-0.5\pi)}A(\Omega)$$

$$\varphi(\Omega) = -\frac{M}{2}\Omega + \frac{\pi}{2}$$

$A(\Omega)$ is odd-symmetric about the point $\Omega = 0$ $\quad A(0)=0$

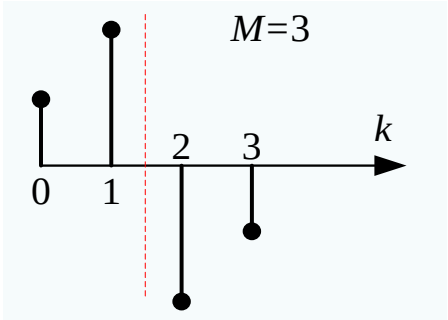
$A(\Omega)$ is odd-symmetric about the point $\Omega = \pi$ $\quad A(\pi)=0$

Cannot be used for the design of low-pass, high-pass, or band-stop filters.

Frequency domain characteristics of linear phase systems

Type IV ($h[k]= -h[M-k]$, M is odd number)

$h[k]=\{1,2,-2,-1\}$, $M=3$



$$H(e^{j\Omega}) = 1 + 2e^{-j\Omega} - 2e^{-j2\Omega} - 1\cdot e^{-j3\Omega}$$

$$= 2e^{-j\frac{3}{2}\Omega}(e^{j\frac{1}{2}\Omega} - e^{-j\frac{1}{2}\Omega}) + e^{-j\frac{3}{2}\Omega}(e^{j\frac{3}{2}\Omega} - e^{-j\frac{3}{2}\Omega})$$

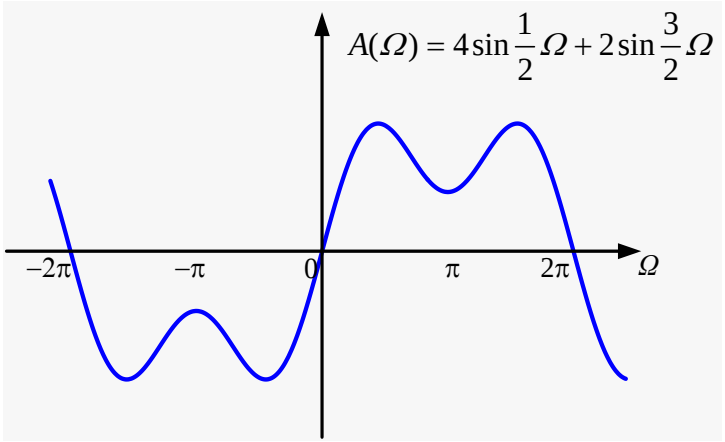
$$= e^{-j(\frac{3}{2}\Omega-\pi/2)}(4\sin\frac{1}{2}\Omega + 2\sin\frac{3}{2}\Omega)$$

$$A(\Omega) = 4\sin\frac{1}{2}\Omega + 2\sin\frac{3}{2}\Omega \qquad \varphi(\Omega) = -\frac{3}{2}\Omega + \pi/2 = -\frac{M}{2}\Omega + \frac{\pi}{2}$$

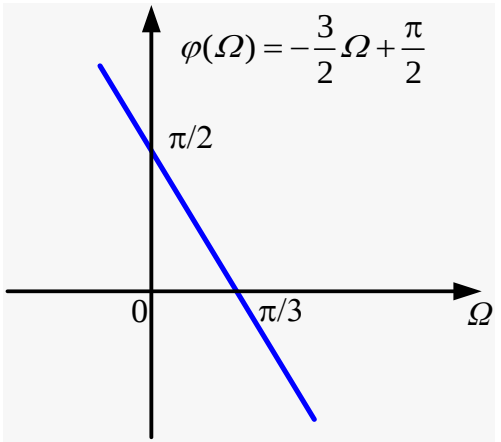
Frequency domain characteristics of linear phase systems

Type IV ($h[k]= -h[M-k]$, M is odd number)

$$H(e^{j\Omega})=e^{-j(\frac{3}{2}\Omega-\pi/2)}(4\sin\frac{1}{2}\Omega+2\sin\frac{3}{2}\Omega)$$



$A(0)=0$



$A(-\Omega)=-A(\Omega)$

$A(2\pi-\Omega)=A(\Omega)$

$A(\Omega)$ is odd-symmetric about the point $\Omega=0$

$A(\Omega)$ is even-symmetric about the point $\Omega=\pi$

Frequency domain characteristics of linear phase systems

Type IV ($h[k]= -h[M-k]$, M is odd number)

$$H(e^{j\Omega}) = e^{-j(0.5M\Omega-0.5\pi)}A(\Omega)$$

$$\varphi(\Omega) = -\frac{M}{2}\Omega + \frac{\pi}{2}$$

$A(\Omega)$ is odd-symmetric about the point $\Omega = 0$

$A(\Omega)$ is even-symmetric about the point $\Omega = \pi$

$$A(0)=0$$

Cannot be used for the design of low-pass filters.

Frequency domain characteristics of linear phase systems

Odd-symmetric: $h[k] = h[M-k]$ (Type I、II)

$$H(e^{j\Omega}) = A(\Omega)e^{-j\frac{M}{2}\Omega} \qquad \varphi(\Omega) = -\frac{M}{2}\Omega$$

Even-symmetric: $h[k] = -h[M-k]$ (Type III、IV)

$$H(e^{j\Omega}) = A(\Omega)e^{-j(\frac{M}{2}\Omega-\frac{\pi}{2})} \qquad \varphi(\Omega) = -\frac{M}{2}\Omega+\frac{\pi}{2}$$

Frequency domain characteristics of linear phase systems

Type	I	II	III	IV
M	even	odd	even	odd
$h[k]$ symmetry	even-symmetric	even-symmetric	odd-symmetric	odd-symmetric
$A(\Omega)$ about $\Omega=0$	even-symmetric	even-symmetric	odd-symmetric	odd-symmetric
$A(\Omega)$about $\Omega=\pi$	even-symmetric	odd-symmetric	odd-symmetric	even-symmetric
$A(0)$	Arbitrary	Arbitrary	0	0
$A(\pi)$	Arbitrary	0	0	Arbitrary
Filter types that are not applicable	–	HP, BS	LP, HP， BS	LP, BS

Zero distribution of linear phase system

The zero distribution characteristics of a linear-phase FIR filter $H(z)$.

$$h[k] = \pm h[M-k] \quad \Longrightarrow \quad H(z) = \pm z^{-M} H(z^{-1})$$

If *h[k]* satisfies even symmetry $h[k] = h[M-k]$ $H(z) = z^{-M} H(z^{-1})$

If *h[k]* satisfies symmetry $h[k] = -h[M-k]$ $H(z) = -z^{-M} H(z^{-1})$

Zero distribution of linear phase system

If z_i is a zero of $H(z)$, $H(z_i)=0$

because $H(z) = \pm z^{-M} H(z^{-1})$

$$\pm z_i^{-M} H(z_i^{-1}) = 0 \Longrightarrow H(z_i^{-1}) = 0$$

$$H(z_i) = 0 \Leftrightarrow H(z_i^{-1}) = 0$$

If z_i is a zero of $H(z)$, then its reciprocal z_i^{-1} is also a zero of $H(z)$

Zero distribution of linear phase system

If z_i is a complex zero of $H(z)$, $H(z_i)=0$

Since the $h[k]$ of a physically realizable FIR system is a real sequence

The complex zeros of $H(z)$ should appear in conjugate pairs.

$$H(z_i)=0 \quad \Leftrightarrow \quad H(z_i^*)=0$$

If z_i is a complex zero of $H(z)$, then its conjugate z_i^* is also a zero of $H(z)$

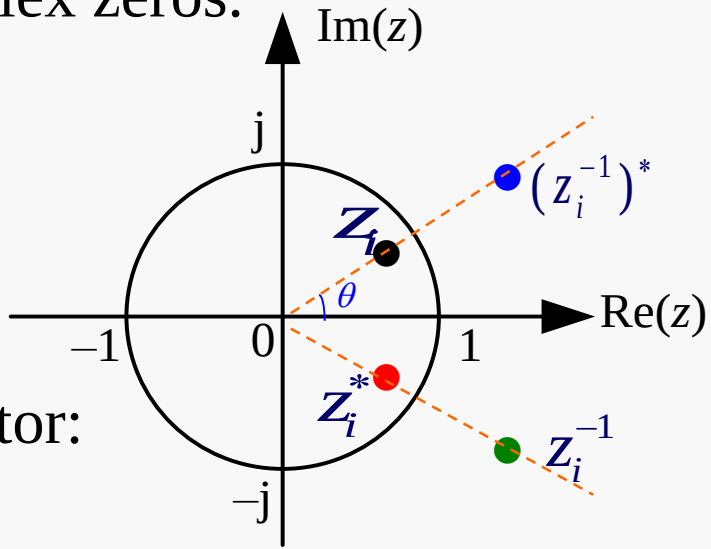
Zero distribution of linear phase system

(1) $z_i = re^{j\theta}$, Complex zeros not on the unit circle ($r \neq 1$, $\theta \neq 0, \pi$)

Then there must be three additional complex zeros.

$$z_i^{-1} = \frac{1}{r}e^{-j\theta} \qquad z_i^* = re^{-j\theta}$$

$$(z_i^{-1})^* = \frac{1}{r}e^{j\theta}$$



Four zeros form a 4th-order z-domain factor:

$$H_1(z) = 1 + az^{-1} + bz^{-2} + az^{-3} + z^{-4}$$

Zero distribution of linear phase system

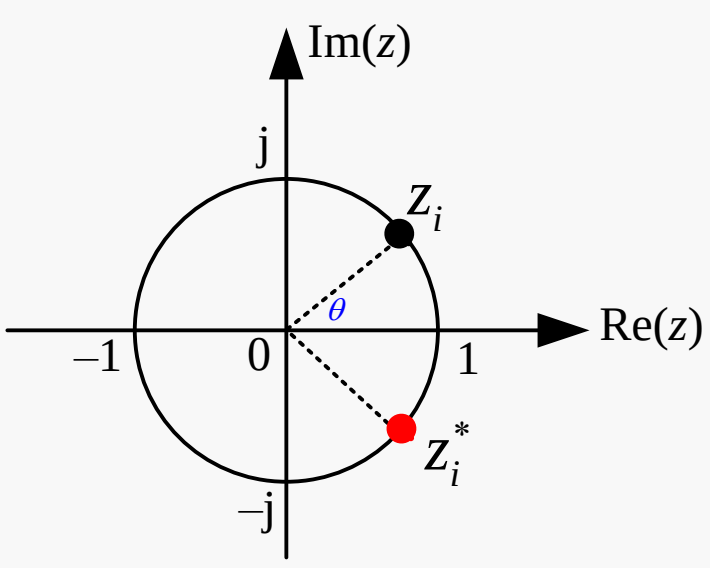
(2) $z_i = re^{j\theta}$, Complex zeros on the unit circle ($r=1$, $\theta \neq 0, \pi$)

$$z_i^{-1} = z_i^* = 1 \cdot e^{-j\theta}$$

$$(z_i^{-1})^* = z_i = 1 \cdot e^{j\theta}$$

z_i and z_i^* form a 2nd-order z-domain factor:

$$H_2(z) = 1 + az^{-1} + z^{-2}$$



Zero distribution of linear phase system

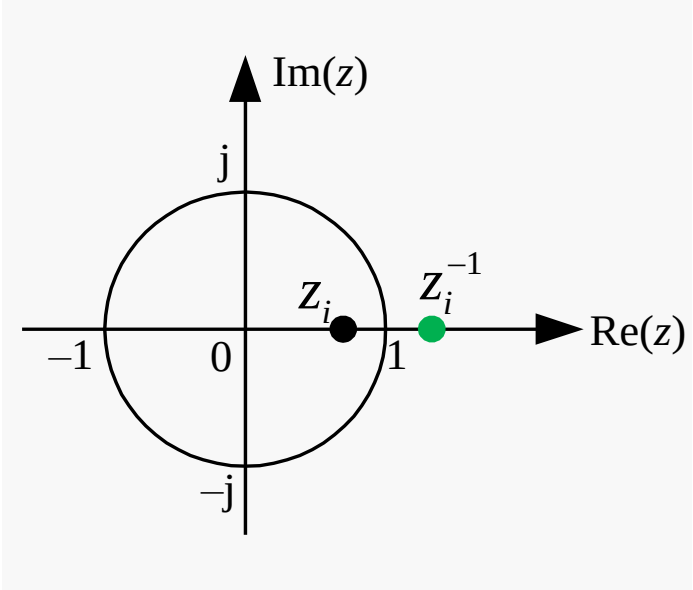
(3) $z_i = re^{j\theta}$, Real zeros not on the unit circle ($r \neq 1$, $\theta = 0, \pi$)

$$z_i = z_i^* = r$$

$$z_i^{-1} = (z_i^{-1})^* = 1/r$$

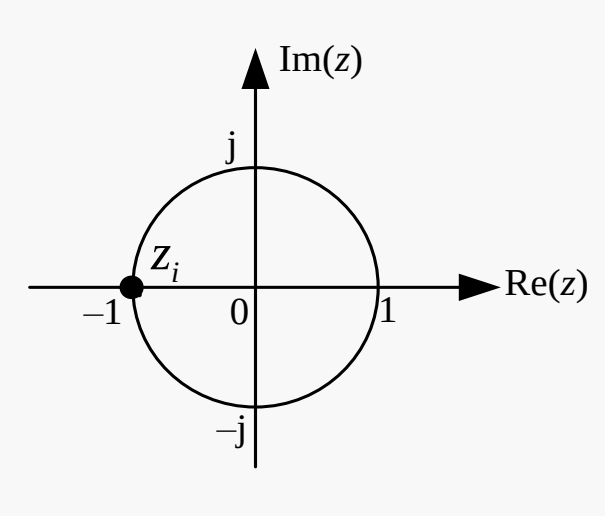
z_i与z_i^{-1} form a 2nd-order z-domain factor

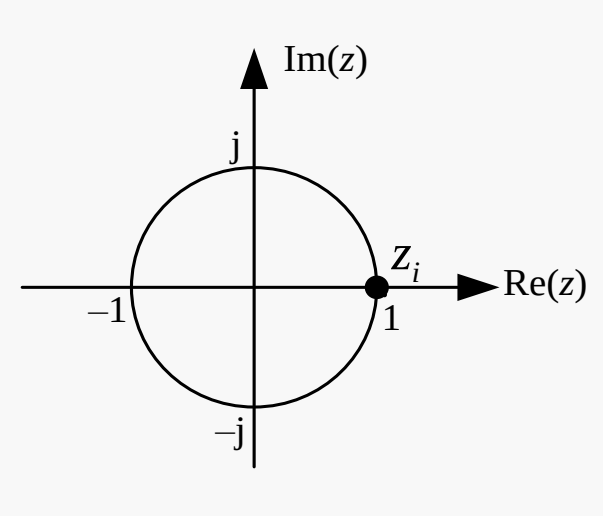
$$H_3(z) = 1 + az^{-1} + z^{-2}$$



Zero distribution of linear phase system

(4) $z_i = re^{j\theta}$, Real zeros on the unit circle ($r=1$, $\theta=0, \pi$)





form a 1st-order z-domain factor: $1+z^{-1}$

form a 1st-order z-domain factor: $1-z^{-1}$

$$H_4(z) = 1 \pm z^{-1}$$

Zero distribution of linear phase system

A linear-phase FIR system is the cascade of the following four sub-systems:

Complex zeros outside the unit circle: $H_1(z) = 1 + az^{-1} + bz^{-2} + az^{-3} + z^{-4}$

Real zeros outside the unit circle: $H_2(z) = 1 + az^{-1} + z^{-2}$ (Two unequal real zeros)

Complex zeros on the unit circle: $H_3(z) = 1 + az^{-1} + z^{-2}$ (Two conjugate complex zeros)

Real zeros on the unit circle: $H_4(z) = 1 \pm z^{-1}$

Zero distribution of linear phase system

Example: The system function of an 8th-order Type III linear-phase FIR filter is:

$$H(z) = 1 - 5.2z^{-1} + 2.2025z^{-2} - 6.253z^{-3} + 6.253z^{-5} - 2.2025z^{-6} + 5.2z^{-7} - z^{-8}$$

Its zeros are:

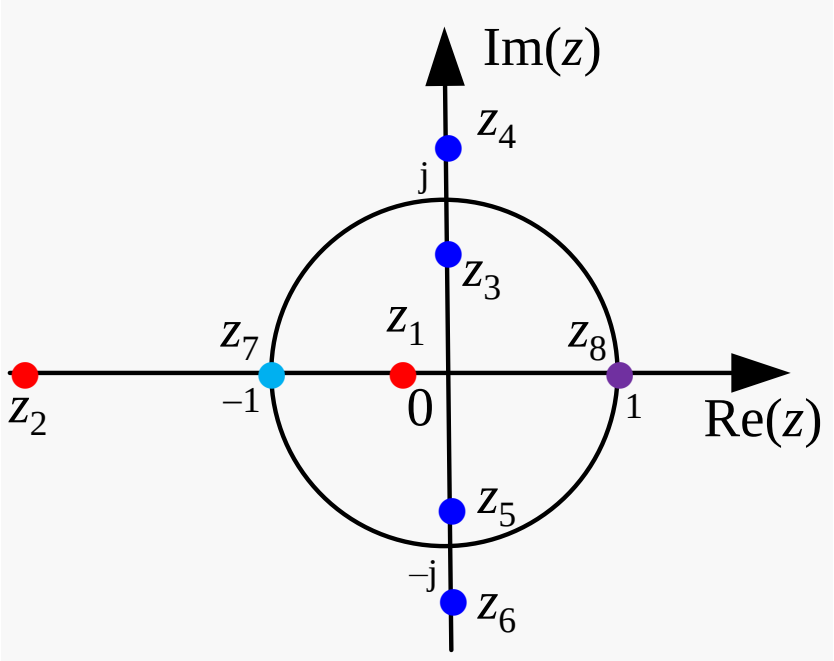
$z_1 = -0.2$, $z_2 = -5$

$z_3 = j0.8$, $z_4 = j1.25$

$z_5 = -j0.8$, $z_6 = -j1.25$

$z_7 = -1$ (Real zeros on the unit circle)

$z_8 = 1$ (Real zeros on the unit circle)



Zero distribution of linear phase system

Example: The system function of an 8th-order Type III linear-phase FIR filter is:

$$H(z)=1-5.2z^{-1}+2.2025z^{-2}-6.253z^{-3}+6.253z^{-5}-2.2025z^{-6}+5.2z^{-7}-z^{-8}$$

Formed by the cascade of the following sub-systems: $H(z)=H_1(z)H_2(z)H_3(z)H_4(z)$

$H_1(z)=1+z^{-1}$ $z_7=-1$ $H_2(z)=1-z^{-1}$ $z_8=1$ 1st

$H_3(z)=1-5.2z^{-1}+z^{-2}$ $z_1=-0.2$, $z_2=-5$ 2nd

$H_4(z)=1+2.2025z^{-2}+z^{-4}$ $z_3=$ j0.8 , $z_4=$ j1.25

$z_5=-$j0.8, $z_6=-$j1.25 4th

Thanks

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude.